

When to Collect Sensitive Category Data? Public Sector Considerations For Balancing Privacy and Freedom from Discrimination in Automated Decision Systems

Anna Lenhart
University of Maryland
alenhart@umd.edu

ABSTRACT

Automated Decision Systems (ADS) are being used to inform important decisions in government services. Concerns regarding discrimination in ADS have led to the rise of bias mitigation techniques, or data science practices that measure and adjust for disparities based on protected class data. These techniques often require demographic or sensitive category data to both measure discrimination in ADS and process data to mitigate the discriminatory bias. The collection of sensitive category data increases privacy concerns. This preliminary study includes nine semi-structured interviews with data practitioners working in government. The analysis explores what considerations data practitioners in the public sector make when determining best practices for sensitive category data collection and how they engage with the tradeoff between privacy and right to freedom from discrimination. Themes pertinent to social services emerged including the importance of accessibility, reasons for data minimization and the role of reporting structures and historical norms.

KEYWORDS

demographic data, sensitive data, public sector, data privacy, fairness, anti-discrimination

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1 BACKGROUND & MOTIVATION

Artificial intelligence, machine learning and rules-based algorithms, a class of data science tools referred to as Automated Decision Systems (ADS) throughout this poster, can be used to make decisions automatically, or influence a human decision. Applications of ADS in the public sector include screening calls related to child protection services [8, 11] or determining whether defendants receive pretrial detention [9, 14]. The use of these applications has led to increased concerns over discrimination as data scientists have

underscored that the data used to train ADS may be embedded with historical patterns of discrimination [1, 12] and/or does not represent the environment in which the ADS will operate [7, 10].

Concerns regarding discrimination have led to the rise of awareness-based techniques for bias mitigation or data science practices that measure and adjust for disparities based on protected class data [5]. Bias mitigation techniques often require sensitive category data such as race, ethnicity, gender, sexual orientation, age, etc to both measure discrimination in ADS and adjust the computer code and data processing to mitigate the discriminatory bias [6, 20]. Collection of sensitive category data puts data subjects at an increased risk of being re-identified, targeted or having records compromised [6, 20].

Recent scholarship argues that data practitioners should address the tradeoff between privacy and freedom from discrimination by considering context and asking questions about where and how to collect sensitive category data [3, 6, 18]. This qualitative study explores how public sector data practitioners, which tend to have less resources and different incentives and norms compared to the private sector, currently consider these tradeoffs.

2 METHODOLOGY & DATA ANALYSIS

The author conducted nine long form (30-60 minutes) semi-structured interviews with individuals working in data administrator and data analysis roles in the public sector (referred to as data practitioners) over Zoom in the Spring of 2021. Participants were recruited using a snowball method, the author used her network from graduate school and former employment working as a data practitioner in the public sector. This method was preferred because public sector data practitioners are a specialized group generally discouraged from speaking publicly about their work. The project had IRB approval from the University of Maryland (1612009-2), identifiers were removed, and participants were spoken about in general terms to protect their anonymity. The protocol was based on demographic data use considerations within existing AI ethics frameworks [19]. The interview was broken down into 4 sections described in Table 1.

The interview transcripts were analyzed using an iterative coding process with one person, the author, coding. There was no inter-rater reliability or group negotiating. The concepts from literature that informed the protocol also acted as the start list for codes. The first round of coding was done on six of the interviews and focused on getting the participants responses clustered by the concepts covered in the protocol. After the first round of coding, the code book definitions were updated, and all nine interviews were

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Table 1: Protocol Outline

Section	Types of Questions
Understanding the Participant	Questions aimed at understanding the participant's roles and responsibilities at the government agencies where they currently or recently served and the details of the data processing projects they contributed to. Examples: •Can you describe your role (current or former) or, in the case of a contractor, a specific project with a governmental body? •What was your relationship like with those who made decisions about when and how to collect data?
Sensitive Category Data Collection	Questions aimed at understanding the types of sensitive category data collected, the manner in which it was collected and how data practitioners think about the data they need and risks to privacy. Examples: •Does your agency follow specific laws/guidelines for the collection (or storage/treatment) of demographic information? •Have you ever chosen not to collect demographic data about individuals in a dataset? If so, why?
Automated Decision Systems	Questions aimed at understanding if and how organizations are using ADS and their ethical considerations. The concepts included: awareness-based techniques for bias mitigation [5], Proxy variables [12], ADS audits [4, 12], interpretation of ADS [4, 17], Contestability [2], Gaming [18] •From your point of view what are the hardest questions your team is grappling with regarding ethical/responsible use of ADS? •What is your team's current approach to ADS auditing, if any?
Considerations for the Future ^a	Directed participants to consider future processes that may help organizations make decisions regarding when to collect sensitive attribute data.

^a This portion of the protocol was not analyzed as part of this preliminary study

Table 2: Participant Description

Category	Example	Number of Participants
Local Government	Data practitioners in city or county governments	5
Federal Government	Data practitioner employed by a federal agency or a government contractor assigned to a federal agency	4
Decision Maker	In a role where the final data collection and processing is determined by the participant	2
Decision Influencer	In an advisory role and able to make suggestions regarding data collection and processing	7

recorded. The number of participants in each level of government and role is described in Table 2.

Participants worked on multiple projects ranging from applications for government services, federal procurement, and human

resources. Participants emphasized that they can only speak from their perspective, not the organizations for which they worked. Table 3 describes the types of projects discussed during the interview. While participants did describe their projects, their answers to many

Table 3: Types of Projects Discussed

Project Type	Decision Systems	Example of Projects	Number of Projects Discussed
Compliance	Rules-based (automated), predictive analytics (automated), Natural Language Processing with rules (automated)	Systems to track a contractor's hours and tasks, or landlord's compliance with affordable housing criteria	4
Information interchange	Rules-based (automated)	Steps in a procurement process, or automatically sending test results	3
Applications for services or funding	Rules-based (automated), human review of raw data (mix of quantitative and qualitative)	Screening for eligibility to receive payments, pensions, funding proposals, services or school admission	4
Program evaluation	No decision system	Randomized control trials, surveys, dashboards to evaluate and justify programs or future programs	5

of the questions in the protocol were informed by their experience on a range of projects. It is worth noting that the author did not speak with anyone working in national defense, criminal justice or a project with risk scoring or facial recognition—projects frequently mentioned in the existing scholarship regarding discrimination in public sector ADS [9, 11, 13].

3 FINDINGS

Four themes emerged as participants discussed considerations regarding best practices for sensitive category data collection and how they engage with the tradeoff between privacy and right to freedom from discrimination.

3.1 Theme 1: Government is focused on ADS as a means to improve processes

Several participants mentioned a focus on using ADS for improving processes. One participant working on federal-level projects explained, “[the government is] very conscientious about making decisions about people. . .there has actually been a focus on prioritizing [ADS] around tasks and processes. . .but not making suggestions about individuals.” Or as one decision maker in local government explains, “I think people get caught up in the shiny stuff and talk about machine learning and data mining and AI as things that they want to do in our in the organization. But there are so many basics we’re not even doing as well as we could.” Another local government participant described, “we want to automate more data collection and cleaning, data transparency and dashboard processes. But in terms of using data and or using AI [for critical] decision making, I don’t think that’s something that’s emerging anytime soon.”

3.2 Theme 2: Discrimination concerns centered on accessibility

When discussing discrimination, participant’s expressed concerns regarding universal accessibility. One participant, focused on health-care provision, explained, “People who need the services most are the ones that are less equipped to handle navigating the bureaucracy.” Another explained the importance of language when designing user-facing systems, both considerations for non-English speakers and using plain language at a reasonable reading level. Lastly, one participant working on a human resources project expressed concerns that users from an older demographic may be less comfortable with the technical system and may be recorded as “out of compliance” when in actuality they struggled to use the application.

3.3 Theme 3: Data minimization was emphasized as a norm for reasons beyond protecting an individual’s privacy

Every participant expressed the importance of data minimization. While many mentioned laws such as Confidential Information Protection and Statistical Efficiency Act (CIPSEA), Health Insurance Portability and Accountability Act of 1996 (HIPAA), and Family Educational Rights and Privacy Act (FERPA), their reasons went beyond legal compliance. A few emphasized the importance of trust in government, one stated, “If we were to allow that data to be compromised, that would irreparably damage our ability to do our work.” Many participants discussed internal practices regarding cell suppression and limiting access to personally identifiable information, one participant working on a project with a high quantity of sensitive data explained, “So it was kind of like a code of honor, I suppose that even though you had access to everything you only use what you really need to answer a question.”

Several practitioners mentioned the importance of not asking for more data than what is needed both to decrease burden on

respondents and on the organization's storage and data management systems. In reference to programs that require applications, one participant said, "I would prefer that we try to minimize the burden on our respondents, and especially questions that might make them uncomfortable." This participant went on to bring up an issue important to public service, stating "I think about setting expectations. . . I try to remind [colleagues] that we don't want to be over promising in our surveys. And we should really only ask questions that are pertinent to what we can do to improve people's wellbeing, and not things that are far outside the scope of what we can do." Similarly, a participant working on a grant application process shared concerns that including questions regarding sensitive category data may communicate that demographic factors would contribute to the outcome of an application, even when it would not. Another participant working on a local level shared that the reliance on spreadsheets added challenges for collecting detailed demographics, "many of our providers don't have sophisticated data systems so collecting additional and more detailed race categories would quickly multiply the number of columns on their spreadsheet."

3.4 Theme 4: The amount and type of sensitive category data collected is informed by reporting structures and historical norms

Participants described the way that many data fields were determined by the other agencies or funders to whom they report. This was especially true for data practitioners working in local government that need to report to federal agencies. For example, when asked about how the options in demographic data drop down lists is determined, one data practitioner explained, "we try to align with other big system actors in the area, many of our programs are reporting to multiple funders, so there are guidelines that we're adhering to." They also emphasized the need for historical comparison, one explained, "for programs that have existed for a very long time, 30, 40, 50, and 60 years, they set the rules for the data that you need to get approved for some program 60 years ago, and they really haven't changed."

4 DISCUSSION & FUTURE WORK

This preliminary study supports existing findings that the availability and norms surrounding sensitive category data collection vary by sector [3, 6, 18]. These preliminary results underscore that data practitioners working in federal agencies (social services) and local government are mostly focused on improving their processes and ensuring that everyone can access services. Additionally, as projects such as those described in this study are included in mandates for algorithm impact assessments, years of data minimization and historical option lists for protected classes may pose challenges for conducting statistical tests or using awareness-based techniques for bias mitigation [5]. Flexibility may be needed to allow for other studies or measures of discrimination in ADS [16] or mandates to ensure efficient contestability processes [2]. The next steps include creating and testing a framework for engaging users, data subjects and key stakeholders in sensitive category data collection policy formulation focused on ADS in social services.

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